

```
In [1]: import os
import sys
from pyspark.sql import SparkSession
from pyspark.sql.functions import avg, col, count

os.environ["PYSPARK_PYTHON"] = sys.executable
os.environ["PYSPARK_DRIVER_PYTHON"] = sys.executable

spark = (
    SparkSession.builder.appName("PySpark_Fix_Test")
    .master("local[*]")
    .getOrCreate()
)

books_data = [
    ("A Light in the Attic", "Poetry", 51.77, 4.5),
    ("Tipping the Velvet", "Fiction", 53.74, 4.0),
    ("Soumission", "Fiction", 50.10, 3.8),
    ("The Requiem Song", "Poetry", 22.65, 4.8),
    ("The Black Cat", "Mystery", 45.11, 4.2),
    ("Starving Hearts", "Fiction", 42.92, 4.1),
    ("Set Boundaries, Find Peace", "Self-Help", 30.00, 4.9),
]

columns = ["Title", "Category", "Price_USD", "Rating"]

df = spark.createDataFrame(books_data, columns)

print("=== 1. Previewing The Dataset ===")
df.show(truncate=False)

print("=== 2. Aggregation: Average Price & Book Count per Category ===")
summary_df = (
    df.groupBy("Category")
    .agg(count("Title").alias("Total_Books"), avg("Price_USD").alias("Avg_Price"))
    .orderBy(col("Avg_Price").desc())
)
```

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summary_df.show()

print("=== 3. Filtered: Top Rated Books (> 4.2) ===")
df.filter(df.Rating > 4.2).select("Title", "Category", "Rating").show(
    truncate=False
)
```

=== 1. Previewing The Dataset ===

Title	Category	Price_USD	Rating
A Light in the Attic	Poetry	51.77	4.5
Tipping the Velvet	Fiction	53.74	4.0
Soumission	Fiction	50.1	3.8
The Requiem Song	Poetry	22.65	4.8
The Black Cat	Mystery	45.11	4.2
Starving Hearts	Fiction	42.92	4.1
Set Boundaries, Find Peace	Self-Help	30.0	4.9

=== 2. Aggregation: Average Price & Book Count per Category ===

Category	Total_Books	Avg_Price
Fiction	3	48.919999999999995
Mystery	1	45.11
Poetry	2	37.21
Self-Help	1	30.0

=== 3. Filtered: Top Rated Books (> 4.2) ===

Title	Category	Rating
A Light in the Attic	Poetry	4.5
The Requiem Song	Poetry	4.8
Set Boundaries, Find Peace	Self-Help	4.9

```
In [2]: import os
import sys
from bs4 import BeautifulSoup
from pyspark.sql import SparkSession
from pyspark.sql.functions import col, count, explode, split
from warcio.archiveiterator import ArchiveIterator

os.environ["PYSPARK_PYTHON"] = sys.executable
os.environ["PYSPARK_DRIVER_PYTHON"] = sys.executable

warc_filename = "quotes_crawl.warc.gz"
extracted_data = []

with open(warc_filename, "rb") as stream:
    for record in ArchiveIterator(stream, check_digests=False):
        if record.rec_type == "response":
            html_content = record.content_stream().read()
            soup = BeautifulSoup(html_content, "html.parser")

            quote_elements = soup.find_all("div", class_="quote")
            for q in quote_elements:
                text = q.find("span", class_="text").text.strip()
                author = q.find("small", class_="author").text.strip()
                tags = [
                    tag.text.strip() for tag in q.find_all("a", class_="tag")
                ]
                extracted_data.append((author, text, ", ".join(tags)))

# PySpark
spark = (
    SparkSession.builder.appName("Real_WARC_Quotes_Analysis")
    .master("local[*]")
    .getOrCreate()
)

columns = ["Author", "Quote", "Tags"]
df_real = spark.createDataFrame(extracted_data, columns)

print(
```

```
f"=== 1. Real Dataset Preview from {warc_filename} (Total: {df_real.count()} quotes) ==="
)
df_real.show(5, truncate=50)

print("=== 2. Real Analytics: Top Authors by Quote Count ===")
author_counts = (
    df_real.groupBy("Author")
    .agg(count("Quote").alias("Total_Quotes"))
    .orderBy(col("Total_Quotes").desc())
)
author_counts.show(truncate=False)

print("=== 3. Real Analytics: Most Popular Tags ===")
top_tags = (
    df_real.withColumn("Single_Tag", explode(split(col("Tags"), ", ")))
    .groupBy("Single_Tag")
    .agg(count("Single_Tag").alias("Tag_Count"))
    .orderBy(col("Tag_Count").desc())
)
top_tags.show(5)
```

=== 1. Real Dataset Preview from quotes_crawl.warc.gz (Total: 10 quotes) ===

Author	Quote	Tags
Albert Einstein	"The world as we have created it is a process o...	change, deep-thoughts, thinking, world
J.K. Rowling	"It is our choices, Harry, that show what we tr...	abilities, choices
Albert Einstein	"There are only two ways to live your life. One...	inspirational, life, live, miracle, miracles
Jane Austen	"The person, be it gentleman or lady, who has n...	aliteracy, books, classic, humor
Marilyn Monroe	"Imperfection is beauty, madness is genius and ...	be-yourself, inspirational

only showing top 5 rows

=== 2. Real Analytics: Top Authors by Quote Count ===

Author	Total_Quotes
Albert Einstein	3
J.K. Rowling	1
Jane Austen	1
Marilyn Monroe	1
André Gide	1
Thomas A. Edison	1
Eleanor Roosevelt	1
Steve Martin	1

=== 3. Real Analytics: Most Popular Tags ===

Single_Tag	Tag_Count
inspirational	3
life	2
humor	2
deep-thoughts	1
world	1

only showing top 5 rows

In []: